Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

PENDAHULUAN
TINJAUAN PUSTAKA
[a4paper,11pt, titlepage, twoside]article
titling ADOreD report template Name Here July 2023
lipsum [T1]fontenc parskip

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graphicx float eurosym amsmath textgreek steinmetz systeme sectsty multirow caption subcaption wrapfig listings [colorlinks=true, allcolors=blue]hyperref color [dvipsnames]xcolor [section]placeins longtable mathtools commath dtk-logos fancyhdr titlesec transparent booktabs graphicx

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Blue Blue RoyalBlue

[sorting=none]biblatex sources.bib

[ER] ADOreD_logo_no_subtext.png [OL] ADOreD report template [ELH, ORH] 1 [L] grayThis project has received funding from the European Union's Horizon Europe
Research and Innovation programme under the Marie Skłodowska-Curie grant agreement No 101073554.

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ADOreD

Accelerating the Deployment of Offshore wind using DC technology

Grant Agreement number : 101073554

Blue

Description of deliverable:

19 Pebruari 2025

Deliverable number	D4.1
Work Package	WP6: Management
Dissemination level	Public
Project period	xxxxxx
Due date	xxxxxx
Actual submission date	xxxxxx

Daftar Isi

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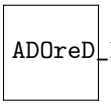
Abstract

1 Introduction

[3-5] [?]

2 Development

[1-2]



Gambar 1: ADOreD Logo

[1-2]

2.1 Helpful Information

[1]

Tabel 1: ADOreD Candidates starting date

Fellow nr	Institution	Name	Starting date - approximately
1	UPC	Kriti Agrawal	1st of September
2	UPC	Bernardo Castro Valerio	19th of June
3	UPC	Luis Angel Garcia-Reyes	1st of July
4	UPORTO	Re-advertisement	? (not possible before October)
5	UPORTO	Waseem Haider	1st of September
6	Clille	Anup Joshi	1st of September
7	KULeuven	Manika Khadka	1st of September
8	YUSO	Re-advertisement	? Contact them
9	DTU	Alexander Novikov	1st of September
10	DTU	Nicolae Darii	1st of July
11	DTU	Germano Mugambe	1st of September
12	DTU	Zhengua Xu (Joshua)	1st of September
13	UPC	Mario Useche Arteaga	1st of July
14	KULeuven	Hanwei Li	1st of September
15	UPORTO	Ahmar Farooque	1st of September
16	Cardiff	Muhammad Fawad	1st of August
17	Cardiff	Qingyuan Gong	1st of August
18	Cardiff	Junaid Khalid	1st of August
19	Cardiff	Muhammad Helal Uddin	1st of August

Tabel 2: Work Package List

WP No.	WP Title	Lead (Short Name)	Start	End	Activity	Reasearchers
1	Offshore Wind	CU	6	48	Research	3, 10, 11, 12, 17, 19
2	HVDC	UPC	6	48	Research	1, 2, 6, 7, 14, 16
3	AC/DC interaction	KULeuven	6	48	Research	4, 5, 8, 9, 13, 15, 18
4	Training	DTU	6	48	Training Activity	all
5	Communication and Dissemination	UPC	6	48	Dissemination	all
6	Management	DTU	1	48+	Management	

Tabel 3: ADOreD projects

Fellow nr	Project title
1	Autonomous operation of modern power systems including HVDC grids
2	Tools for the design and sizing of offshore transmission systems using HVDC systems and grids
3	Tools development for stability and interaction assessment of large modern power electronics dominated power systems.
4	Hydrogen storage on offshore wind power plants
5	Digitalization of the offshore power substation secondary equipment in an AC/DC context
6	Stability of an islanded power system with a grid-forming VSC-HVDC link, an LCC-HVDC link and non-synchronous generation
7	HVDC Cable Analytics: Using monitored data from contingency analysis to determine cable and accessory lifetime
8	Risk-based operational models for hybrid offshore interconnectors for ancillary service provision
9	Wind turbine and wind power plant protection in a converter-based power system
10	Electrical Compatibility Study of a Converter based Offshore Energy System constituting diverse Control Strategies.
11	Dynamic stability of multi-vendor large (offshore) wind power clusters
12	Ancillary services from offshore wind power plants in HVDC grids
13	Tools development for stability and interaction assessment of large modern power electronics dominated power systems
14	Protection schemes for HVDC cable systems in multi-vendor HVDC grids
15	Renewable Hydrogen on offshore wind power plants
16	Assessment and operation of DC technology based multi-purpose interconnectors (MPIs)
17	Design and control of offshore wind farms with hybrid ac and dc technologies
18	Modelling and control of large wind turbines for fault ride-through and grid support
19	Reliability-based design optimisation of large offshore wind turbine

2.2 How to add Lists

You can make lists with automatic numbering ...

1. Like this,
2. and like this.

...or bullet points ...

- Like this,
- and like this.

2.3 How to write Mathematics

LaTeX is great at typesetting mathematics. Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed random variables with $E[X_i] = \mu$ and $Var[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{1}{n}\sum_{i}^{n} X_i \tag{1}$$

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$.

2.4 Python program example

[language=Python,style=Python] def create$_Y bus(self) : offdiagonalelements for kinrange(self.nl) : line = self.lines[k] fromNode = line.fromNode.nodeNumbertoNode = line.toNode.nodeNumber$

```
    self.Ybus[fromNode, toNode] -= 1/line.Z self.Ybus[toNode, fromNode] = self.Ybus[fromNode, toNode]

    self.AdmitanceVec[fromNode] += line.Y/2 self.AdmitanceVec[toNode] += line.Y/2
```

2.5 Matlab program example

[language=Matlab,style=Matlab] function [$\mathtt{Y}_bus$] = $Ybus(Line_data_table_pu, total_nodes)$

$\mathtt{Y}_bus = [zeros(total_nodes)];$

$\mathtt{admitance}_vector = [zeros(total_nodes, 1)];$

for i=1:$\mathtt{total}_nodes - 1$

$\mathtt{Y}_bus(k, m) = -1/(R + 1j * X); Y_bus(m, k) = Y_bus(k, m);$

$\mathtt{admitance}_vector(k) = admitance_vector(k)+1j*B/2; admitance_vector(m) = admitance_vector(m)+1j*B/2; end$

for m=1:$\mathtt{total}_nodessum_pa = -sum(Y_bus(:, m)); Y_bus(m, m) = sum_pa + admitance_vector(m); end$

end

3 Referencing

To add references download the bibtex citation style, and copy and paste it to sources.bib such as:

```
    @article{ShortName,
title=The Article,
author=Shall Not be Named,
journal=The Journal,
volume=56,
number=1,
pages=99-111,
year=3662,
publisher=ADored Publisher
}
```

The ´ShortName´ is a personal tag for each citation. To cite on the text use command "\cite{ShortName}"[?]

Conclusion

[3-5]

Annex